M1b Report: Two Anomalies, Minimally Reproduced

Report v0.1, Project secret3, Row A infrastructure, August 11, 2026. Environment: Idris 2, version 0.8.0-1b5176dc9, Ubuntu sandbox, 4 GB.

1. Anomaly A: the coverage knife-edge

A total, two-clause function over a two-constructor indexed family is rejected as not covering. The shipped thirteen-line file CoverageRepro.idr fires it deterministically; check it with one command and read "h is not covering" over clauses that name both constructors. The bisection matrix, twenty-two variants preserved in the project logs, locates a knife-edge: the failure requires, jointly, one constructor whose type index repeats a variable (Arr a (Arr b a)), a second constructor pinned at a distinct constant of the other index (CS against CK), and that second constructor's type index at depth three (the S-axiom's arrow shape). Remove any one ingredient and coverage passes: linearize the repeated variable, pass; pin both constructors at the same constant, pass; shorten the deep index by one arrow, pass; replace the constant partner by a variable-headed one, pass; drop the second index entirely, pass, even with two nonlinear constructors. The production instance is any Hilbert-style derivation family, the K and S axioms with modus ponens, which is how the anomaly first surfaced, as redDeriv in the garden campaign, and, in hindsight, as the EyeOne failures. Upstream candidate located by title through the search API, idris-lang Idris2 issue 402, open, "Covering test fails for multiple linear non-trivial data types"; the body was not retrievable under rate limits, so the correspondence is asserted at title level only. **Workaround, proven in production: Church-encode the family.** GardenSN's DedC carries the derivations as their own recursors; consuming one is application, never matching, and the entire strong-normalization campaign closed on it.

2. Anomaly B: the elaboration space blowup

One project part, the reducibility mutual of the strong-normalization proof, elaborates inside its original 3,600-line monolith within the 4 GB sandbox, on the historical record, and exceeds 12 GB when elaborated in any smaller context: a package module with minimal imports, a 340-line dependency slice, and the monolith's own 2,229-line warm prefix, all killed, under both swap regimes and under tight garbage collection. The shipped MemoryRepro.idr, 244 lines, Core essentials plus the part verbatim, reproduces the kill with one command. Internal bisection localizes the cost to the final mutual block, twenty-one deep dependent clauses over the Starling's three-way duplication; the blocks before it elaborate in seconds. Upstream candidate, same caveat as above: issue 2341, open, "Space-leak occurs when non-trivial dependently typed function is defined via pattern match"; the context-variance, cheap inside a large file and explosive outside it, is offered as a new data point. **Workaround, proven in production: restyle deep dependent mutuals into view-plus-closure machines.** GardenSN replaces the block with four inversion views and two closures, satRed and satCol, and the identical mathematics elaborates in seconds.

3. Disciplines codified

Two lines join the checker disciplines. Church-encode what the checker will not split. And restyle what the elaborator will not afford: head-inversion views first, then small closures over them, never

twenty-one-clause dependent walls. Both disciplines earned their receipts the same night they were named.

4. Status

M1b closes as documented and upstream-linked. The two repro files ship beside this report; filing the issues upstream needs an identity and is left to the author, with the files ready to paste. Nothing in the program remains blocked by either anomaly: coverage routed through DedC, space routed through satRed, and part S's content re-proved where every campaign can reach it.